

1. Overview

Each AI player controls one or more territories. On every AI turn, each territory is processed independently through the decision loop. The loop runs up to 100 iterations per territory; it stops as soon as no action is taken in a given iteration.

All affordability checks require both of the following to be true:

- Current territory balance $\geq$ action cost
- Current territory balance + (Income – (current upkeep + new entity upkeep)) ≥ 0

This prevents the AI from taking actions that would leave a territory cash-negative or in a permanently losing income/upkeep ratio.

2. Difficulty Modifiers

Two independent systems are affected by difficulty:

Difficulty	Action skip chance (per iteration)	Income modifier (end-of-turn, per owned tile)
easy	40%	0
medium	20%	0
hard	0%	0
super_hard	0%	+1 gold per tile

The income modifier is applied at end-of-turn (after the AI decision loop). super_hard AI territories accumulate considerably more gold over time.

3. First-Turn Free Tower Placement

Before any decision-loop actions on turn 1, each AI territory with ≥ 2 tiles receives one free tower at no cost.

Placement rules:

- Eligible tiles: not mountain, not lake, not already occupied, not in graveyard.
- The tile whose 6 hex-neighbours contain the most AI-owned tiles is chosen.
- Each territory may only receive one free tower per game (tracked per owner).

After the free-tower phase, the normal decision loop is skipped for turn 1 entirely. The AI does not buy, move, or build anything else on turn 1.

4. Territory State: ATTACKING vs. DEFENDING

At the start of each decision-loop iteration the state is recomputed:

- DEFENDING: an enemy unit exists within hex-distance 3 of any border tile, belongs to a territory that borders the AI, and has strength $>$ the AI territory's own maximum unit strength.

- ATTACKING: otherwise.

The state is published to the UI each time it is set (colour overlay in developer mode).

5. Decision Priority Chain

Priority 1 (DEFENDING) — Split or Reduce the Stronger Enemy

Goal: weaken the threatening enemy by capturing a tile that splits it or bankrupts it.

Priority 1A

- Hvis området har en unit der kan splitte den stærkere modstanders område, så gør det.
- Prioritet: Lav område med den stærkere unit på så lille, at det område går bankerot i næste runde. Ellers lav området så lille som muligt.

Priority 1B

- Hvis området har to units der kan merge og derefter splitte den stærkere modstanders område, så gør det.
- Prioritet: Lav område med den stærkere unit på så lille, at det område går bankerot i næste runde. Ellers lav området så lille som muligt.

Priority 1C

- Hvis området kan købe en unit der kan splitte den stærkere modstanders område, så gør det.
- Prioritet: Lav område med den stærkere unit på så lille, at det område går bankerot i næste runde. Ellers lav området så lille som muligt.

Priority 2 (DEFENDING) — Reinforce or Build Defense

Goal: raise the territory's defensive capability to match the enemy's strength.

Priority 2A

- Scan all non-unit entities in the territory that are within a 5 hex radius of the stronger enemy unit. If it has an upgrade path and upgraded strength = enemy strength, and is affordable, execute the upgrade.

Priority 2B

- Scan all unit entities in the territory that are within a 5 hex radius of the stronger enemy unit. If its strength = enemy strength, move the unit towards the stronger enemy unit.

Priority 2C

- Scan all unit entities in the territory, that are within a 5 hex radius of the stronger enemy unit. If it has an upgrade path and upgraded strength = enemy strength, and is affordable, execute the upgrade and move the unit towards the stronger enemy unit.

Priority 2D

- Build a tower or castle (strength = enemy strength) as close to the stronger enemy unit as possible.
- Placement filtered by dtSpacedPlacements (≥ 3 tile spacing from existing towers/castles, fallback to ≥ 2).
- Priority: Border tile -1 $\rightarrow$ Border tile.

Priority 2E

- Buy a new unit (strength = enemy strength) as close to the stronger enemy unit as possible.

Priority A — Split Any Enemy Territory

Goal: weaken the enemy (Player + AI) by capturing a tile that splits it.

Priority A1

- Hvis området har en unit der kan splitte en modstanders område, så gør det.
- Prioritet: Opdel et område så modstanders område bliver så lille at det går bankerot i den efterfølgende runde → Brug 1-hex rule hvis det fjerner units eller bygninger → Opdel området.

Priority A2

- Hvis området har to units der kan merge og derefter kan splitte en modstanders område, så gør det.
- Prioritet: Opdel et område så modstanders område bliver så lille at det går bankerot i den efterfølgende runde → Brug 1-hex rule hvis det fjerner units eller bygninger → Opdel området.

Priority A3

- Hvis området kan købe en unit der kan splitte en modstanders område, så gør det.
- Prioritet: Opdel et område så modstanders område bliver så lille at det går bankerot i den efterfølgende runde → Brug 1-hex rule hvis det fjerner units eller bygninger → Opdel området.

Priority B — Connect Own Fragmented Territories (Land Bridge)

Identify a neutral or enemy land tile that sits between the current territory and another AI-owned territory (1-step or 2-step land/water bridge). Sort by size of the territory being connected to (largest first). Try in order:

Priority B1

- Move an existing unit (strength > ZoC) onto the land bridge tile.

Priority B2

- Merge two units so the merged unit can reach the land bridge tile.

Priority B3

- Buy a new unit directly on the land bridge tile (outside-buy), choosing the weakest type that beats ZoC.

Priority B4

- Buy a water bridge if the tile is a Lake and if affordable (cost 5, upkeep 1/turn).

Priority C — Defend Undefined Border Tiles

Triggers when a border tile is empty (or only has a rebel) and is adjacent to an enemy territory with ≥ 2 tiles.

Placement preference: interior tiles adjacent to the undefined border tile (not the border tile itself), filtered through `dtSpacedPlacements`. Falls back to the border tile if no interior tile qualifies.

Tries tower first, then castle (castle requires an `advanced_unit` or `expert_unit` in the territory).

Affordable only.

Priority D — Build a City

Conditions:

- Territory has ≥ 6 tiles.
- Territory does not already contain a city.
- Can afford the city cost (no ongoing upkeep).
- Candidate tile must be within the zone-of-control of an existing AI tower or castle (i.e., the tower/castle tile itself or one of its 6 immediate neighbours).

Among qualifying tiles, the one farthest from the largest enemy territory's representative tile is chosen (cities are built away from the front line).

Priority E1 — Destroy enemy entity

Priority E1a

- Move a unit onto a reachable enemy tile that has a tower, castle, or unit, strength $> \text{ZoC}$.
- Sort: castle $\rightarrow$ tower $\rightarrow$ advanced unit $\rightarrow$ simple unit.

Priority E1b

- Merge 2 units and move the new unit onto a reachable enemy tile that has a tower, castle, or unit, strength $> \text{ZoC}$.
- Sort: castle $\rightarrow$ tower $\rightarrow$ advanced unit $\rightarrow$ simple unit.

Priority E1c

- Buy (outside-buy) a unit (strength $> \text{ZoC}$) that can capture an enemy tile that has a tower, castle, or unit, strength $> \text{ZoC}$. Tries simple unit $\rightarrow$ advanced unit $\rightarrow$ expert unit.
- Sort: castle $\rightarrow$ tower $\rightarrow$ advanced unit $\rightarrow$ simple unit.

Priority E2 — Destroy enemy entity, with Water Bridge

Priority E2a

- Conditions:
 - Find a lake tile adjacent to the current territory with no bridge. The lake tile must be adjacent to a reachable enemy tile that has a tower, castle, or unit.
 - At least one available unit must be adjacent to that lake tile AND strong enough to capture one of those targets this same round (ZoC check).
 - If all conditions met, build the bridge.
- Move a unit onto a reachable enemy tile that has a tower, castle, or unit, strength $> \text{ZoC}$.
- Sort: castle $\rightarrow$ tower $\rightarrow$ advanced unit $\rightarrow$ simple unit.

Priority E2b

- Conditions:
 - Find a lake tile adjacent to the current territory with no bridge. The lake tile must be adjacent to a reachable enemy tile that has a tower, castle, or unit.
 - At least one available unit must be adjacent to that lake tile AND strong enough to capture one of those targets this same round (ZoC check).
 - If all conditions met, build the bridge.
- Merge 2 units and move the new unit onto a reachable enemy tile that has a tower, castle, or unit, strength $> \text{ZoC}$.

- Sort: castle → tower → advanced unit → simple unit.

Priority E2c

- Conditions:
 - Find a lake tile adjacent to the current territory with no bridge. The lake tile must be adjacent to a reachable enemy tile that has a tower, castle, or unit.
 - At least one available unit must be adjacent to that lake tile AND strong enough to capture one of those targets this same round (ZoC check).
 - If all conditions met, build the bridge.
- Buy (outside-buy) a unit (strength > ZoC) that can capture an enemy tile that has a tower, castle, or unit, strength > ZoC. Tries simple unit → advanced unit → expert unit.
- Sort: castle → tower → advanced unit → simple unit.

Priority F — Clear Rebel Tiles (+25%)

If the territory is larger than 12+ tiles check if more than 25% are under Rebel control.

- Step 1: move the weakest available interior unit (not adjacent to any enemy) onto each rebel tile.
- Step 2: if no unit can reach, buy a simple_unit directly on the rebel tile.

Priority G — Remove unaffordable units

If the territory's Current territory balance + (Income – current upkeep) < 0, remove the most expensive unit in the territory.

Priority H1 — Border Expansion vs. enemy

Priority H1a

- Move onto an enemy tile with no non-rebel entity.
- Sort: tiles with Void/Mountain as neighbour → tiles from smaller enemy territory → tiles from larger enemy territory

Priority H2b

- Merge 2 units and move onto an enemy tile with no non-rebel entity.
- Sort: tiles with Void/Mountain as neighbour → tiles from smaller enemy territory → tiles from larger enemy territory

Priority H3c

- Buy (outside-buy) a unit (strength > ZoC) that can capture an enemy tile with no non-rebel entity. Tries simple unit → advanced unit → expert unit.
- Sort: tiles with Void/Mountain as neighbour → tiles from smaller enemy territory → tiles from larger enemy territory

Priority I2 — Border Expansion vs. enemy with Water Bridge

Priority I2a

- Conditions:

- Find a lake tile adjacent to the current territory with no bridge. The lake tile must be adjacent to a reachable enemy tile that has a tower, castle, or unit.
- At least one available unit must be adjacent to that lake tile AND strong enough to capture one of those targets this same round (ZoC check).
- If all conditions met, build the bridge.
- Move onto an enemy tile with no non-rebel entity.
- Sort: tiles with Void/Mountain as neighbour → tiles from smaller enemy territory → tiles from larger enemy territory

Priority I2b

- Conditions:
 - Find a lake tile adjacent to the current territory with no bridge. The lake tile must be adjacent to a reachable enemy tile that has a tower, castle, or unit.
 - At least one available unit must be adjacent to that lake tile AND strong enough to capture one of those targets this same round (ZoC check).
 - If all conditions met, build the bridge.
- Merge 2 units and move onto an enemy tile with no non-rebel entity.
- Sort: tiles with Void/Mountain as neighbour → tiles from smaller enemy territory → tiles from larger enemy territory

Priority I2c

- Conditions:
 - Find a lake tile adjacent to the current territory with no bridge. The lake tile must be adjacent to a reachable enemy tile that has a tower, castle, or unit.
 - At least one available unit must be adjacent to that lake tile AND strong enough to capture one of those targets this same round (ZoC check).
 - If all conditions met, build the bridge.
- Buy (outside-buy) a unit (strength > ZoC) that can capture an enemy tile with no non-rebel entity. Tries simple unit → advanced unit → expert unit.
- Sort: tiles with Void/Mountain as neighbour → tiles from smaller enemy territory → tiles from larger enemy territory

Priority J — Clear Rebel Tiles (+10%)

If the territory is larger than 12+ tiles check if more than 10% are under Rebel control.

- Step 1: move the weakest available interior unit (not adjacent to any enemy) onto each rebel tile.
- Step 2: if no unit can reach, buy a simple_unit directly on the rebel tile.

Priority K — Border Expansion vs. neutral

Priority K1

- Move onto a neutral tile.
- Priority: city tile → grass → forest → desert.

Priority K2

- Merge 2 units and move onto a neutral tile.

- Priority: city tile → grass → forest → desert.

Priority K3

- Buy (outside-buy) a simple unit that can capture a neutral tile.
- Priority: city tile → grass → forest → desert.

Priority L — Clear Rebel Tiles

If the territory is larger than 6+ tiles.

- Step 1: move the weakest available interior unit (not adjacent to any enemy) onto each rebel tile.
- Step 2: if no unit can reach, buy a simple_unit directly on the rebel tile, provided balance covers the cost and the reclaimed tile income offsets the upkeep.

Priority M — Move Unit Closer to Enemy

For each unspent unit that is not already on a border tile: move it to the reachable position that minimises the minimum hex-distance to any enemy tile (mountains excluded from enemy tile list).

Priority N — Remove Unnecessary Units and Buildings

- If the territory is smaller than 4 tiles remove any units.
- For each tower or castle in the territory: if no enemy territory (2+ tile) exists within hex-distance 6, remove the building to reduce upkeep.
- For each bridge on a lake tile in the territory: if 5 or more of the 6 neighbouring tiles are AI-owned, the bridge is no longer useful as a crossing — remove it.